

23ME601	INDUSTRIAL SAFETY	L	T	P	C
		3	0	0	3
Category	Open Elective Courses (OEC)				
Pre requisites					

Course Objectives

The course is intended to

- To enable students to understand the basic Industrial safety engineering acts and rules.
- To impart knowledge on OSHAS (Occupational Safety and Health Assessment Series) in engineering Industry.
- To develop skills in managing, investigating, and optimizing workplace safety.
- To enable the students to identify the causes of accidents and its preventions.
- To train students to identify hazard and assess the risks using suitable techniques

Course Outcomes (COs)

On successful completion of the course, students will be able to

CO. No	Course Outcome	Bloom's Level
CO 1	Identify the evolution of industrial safety acts, rules and health standards.	Remembering
CO 2	Summarize different safety management activities in industry.	Understanding
CO 3	Develop accident investigation reports and create preventive guidelines for industry	Creating
CO 4	Analyze the process to avoid, prevent and control workplace hazards	Analyzing
CO 5	Analyze the role of government agencies and private consulting agencies in safety training.	Analyzing

Course Contents		
Unit – I	BASICS OF SAFETY ENGINEERING & ACTS	9
Evolution of modern safety concept – safety audit; Acts– factories act– 1948–statutory authorities– inspecting staff–Tamil Nadu factories Rules 1950 under safety and health – environment act 1986 – air act 1981, water act 1974 – labour laws; safety in industries – general safety concepts, machine guarding, hazards in metal removing process, Hazardous Wastes (management, handling and Transboundary Movement) Rules 2016, check list for LPG installations, safety precautions using CNG. Introduction to OHSAS 18000 and 14000, National Disaster Management Act.		
Unit – II	SAFETY MANAGEMENT	9
History of Safety movement – general concepts of management – planning for safety for optimization of productivity -productivity, quality and safety-line staff functions for safety -budgeting for safety - safety policy. Incident Recall Technique (IRT), disaster control, job safety analysis, safety survey, safety inspection, safety sampling. Fire Explosion and toxicity Index.		
Unit – III	ACCIDENT INVESTIGATION AND REPORTING	9
Concept of an accident, reportable and non-reportable accidents, unsafe act and condition – principles of accident prevention, Supervisory role- Role of safety committee – Accident causation models - Cost of accident. Overall accident investigation process-Response to accidents, India reporting requirement, Planning document, Planning matrix, Investigators Kit, functions of investigator, four types of evidences, Records of accidents - Class exercise with case study.		
Unit – IV	SAFETY PERFORMANCE MONITORING:	9
Reactive and proactive monitoring techniques - Permanent total disabilities, permanent partial disabilities, temporary total disabilities Calculation of accident indices, frequency rate, severity rate, frequency severity incidence, incident rate, accident rate, safety “t” score, safety activity rate		
Unit – V	SAFETY EDUCATION AND TRAINING	9
Importance of training – identification of training needs – training methods – programme, seminars, conferences, competitions – method of promoting safe practice - motivation – communication - role of government agencies and private consulting agencies in safety training– creating awareness, awards, celebrations, safety posters, safety displays, safety pledge, safety incentive scheme, safety campaign – Domestic Safety and Training, Evaluating Employee’s training Performance		
Total :45 Periods		

Text Books
1. Charles D. Reese “Occupational Health and Safety Management: A Practical Approach”, 3rd Edition CRC press 2017
2. Mark A. Friend, James P. Kohn “Fundamentals of Occupational Safety and Health” 6 th Edition Bernan press, 2016.
3. Krishnan N.V., “Safety Management in Industry”, Jaico Publishing House, Bombay, 2016.

Reference Books

1. Joel M. Haight, "Principles of Industrial Safety", ASSE publishers, 2017
2. R.K.Mishra, "Safety Management", AITBS publishers,2022
3. Relevant India Acts and Rules, Government of India, 2020.
4. C. Ray Asfahl, David W. Rieske "Industrial Safety and health management",Practice,7th Edition, Pearson,2021

Mapping of Course Outcomes (COs) with Programme Outcomes (POs) Programme Specific Outcomes (PSOs)

COs	POs												PSOs	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
CO 1	2	2	2			3		2			2			
CO 2	2	2	2			3		2			2			
CO 3	2	2	2			3		2			2			
CO 4	2	2	2			3		2			2			
CO 5	2	2	2			3		2			2			
Average	2	2	2			3		2			2			

"3"– High
Correlation

"2" – Medium

"1" – low

"-" - No